# Connector Economies
# 变量构造目录与使用指南

内部严格审查材料

variable_catalog_v0.3.0

由 F:/working_paper_wenxiao/trade_welfare/docs/variable_construction_catalog.md
可重复生成

用途：合作者复核数据来源、处理规则、变量定义、验证结果和解释边界。

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# Material Passport

- Origin Skill: stata-latex-research
- Origin Date: 2026-07-27
- Verification Status: VERIFIED_THROUGH_M3.2
- Version Label: variable_catalog_v0.3.0
- Current Data Stage: M0-M3.2 completed
- Next Construction Stage: M3.3 not started
- Canonical Format: Parquet
- Collaborator Format: Stata .dta

# 0. 结论和使用方式

从 M3.1 开始，变量构造应以本目录为变量合同。它回答六个问题：

1. 变量在哪一层数据上构造；
2. 变量的公式、分母、时间窗口和单位是什么；
3. 变量用于 connector ranking、rerouting、production validation 还是 structural moments；
4. 零值、缺失、负值、small base、suppression 和 confidentiality 如何处理；
5. 变量能否求和、能否跨 connector 比较、能否进入回归或模型；
6. 导出 Stata 时使用什么名称、storage type、主键和验证命令。

本目录同时包含三类变量：

- EXISTING_VALIDATED：M2、M3.1 或 M3.2 已存在且 validation PASS；
- PLANNED_M3 至 PLANNED_M6：已经定义但尚未构造，必须在对应阶段完成后补充实际 output hash 和 validation；
- STRUCTURAL_TARGET_ONLY：只能作为模型校准目标或输入候选，不能直接解释为已经识别的结构参数。

凡是本目录未登记的正式变量，不应直接写入最终 processed data。临时探索变量可以存在于 tmp/ 或交互式 Stata session，但进入正式输出前必须补齐本目录。

本目录与 docs/data_processing_audit_trail.md 构成同一套审查材料。本目录不重复数据来源和完整处理历史，而是引用审计档案中的输入、输出和 validation 证据；审计档案也不能替代这里的变量级公式、用途和解释边界。M3-M7 任一阶段完成时，两份 Markdown、两份 PDF 和对应 build summary 必须同步更新。

# 1. 状态、粒度和可加总性代码

## 1.1 状态

| 状态 | 含义 |
|---|---|
| EXISTING_VALIDATED | 已在 processed Parquet 中存在，主键和 validation 已通过 |
| PLANNED_M3 | M3.3-M3.6 的 small-base、replacement、ranking 和 sensitivity inputs 尚未构造 |
| PLANNED_M4 | production-side standardized long table 时构造 |
| PLANNED_M5 | trade 和 production evidence 合并时构造 |
| PLANNED_M6 | ranking、rerouting、validation 和 structural moments 时构造 |
| STRUCTURAL_TARGET_ONLY | 观测 moment 或 candidate shock，不等于已识别结构参数 |
| PROHIBITED | 明确禁止构造或禁止解释的变量 |

## 1.2 粒度代码

| Grain ID | 唯一键 | 主要文件 |
|---|---|---|
| G_PRODUCT_YEAR | analysis_hs6, year | product treatment |
| G_CONN_PRODUCT_YEAR | connector_iso3, analysis_hs6, year | connector trade |
| G_ORIGIN_PRODUCT_MONTH | origin_iso3, analysis_hs6, year, month | Census monthly |
| G_CONN_SECTOR_YEAR | connector_iso3, analysis_sector, year | M3 trade-sector |
| G_CONN_SECTOR_WINDOW | connector_iso3, analysis_sector, window_id | M3 window summaries |
| G_CONN_SECTOR_COMPARISON | connector_iso3, analysis_sector | M3 pre/shock/post comparison and growth |
| G_PROD_LONG | country_iso3, analysis_sector, year, source_name, metric_name，必要时增加 direction/parent | M4 production long |
| G_EVIDENCE | connector_iso3, analysis_sector, year | M5 integrated evidence |
| G_RANK | connector_iso3, analysis_sector, ranking_spec | M6 ranking |
| G_MOMENT | moment_id, scenario | M6 structural moments |

## 1.3 可加总性代码

| Code | 规则 |
|---|---|
| ADD_WITHIN_KEYS | 可以在不跨越声明维度时求和，例如产品金额聚合到 sector |
| NONADDITIVE_RATE | rate/share/exposure 不能直接求和，必须重算分子分母 |
| NONADDITIVE_REPEAT | 为比较而重复，绝不能跨重复维度求和 |
| STOCK | 时点/期末 stock，不能跨年求和 |
| FLOW | 期间 flow，可在不重叠期间内求和 |
| LEVEL | level，可比较或求差，但通常不跨单位/来源相加 |
| FLAG | 逻辑状态，不作金额加总 |

# 2. 通用键、标签和 provenance 变量

这些字段必须尽可能贯穿 M3-M7。它们不是辅助杂项，而是合作者复核 join 和解释边界的基础。

| Canonical name | Stata name | Type | Grain | 用途 | 状态 |
|---|---|---|---|---|---|
| connector_iso3 | conn_iso3 | str3 | connector grains | connector 国家键 | existing/planned |
| country_iso3 | country_iso3 | str3 | production | production host/country 键 | planned M4 |
| analysis_hs6 | hs6 | str6 | product grains | 保留 leading zeros 的产品键 | existing |
| analysis_sector | ana_sector | str32 | sector grains | 第一阶段共同经验 sector | existing |
| year | year | int | annual grains | calendar year | existing |
| month | month | byte | monthly | 1-12 | existing |
| window_id | window_id | str8 | window | pre/shock/post | existing validated |
| source_name | source | str20 | production long | 数据来源区分 | planned M4 |
| metric_name | metric | str32 | production long | 指标区分 | planned M4 |
| unit | unit | str32 | all long outputs | 单位，禁止跨单位相加 | planned M4 |
| mapping_version | map_ver | str32 | all | concordance 版本 | planned |
| build_version | build_ver | str32 | all | builder/output schema 版本 | existing/planned |
| source_row_count | src_n | long | all | 聚合 cell 的原始行数 | existing/planned |
| source_file_count | srcfile_n | int | all | provenance 完整性 | existing/planned |
| mapping_status | map_status | str32 | all | mapped/aggregate/residual/ambiguous 等 | existing/planned |
| is_analysis_eligible | ana_ok | byte | all | 是否进入第一阶段主分析 | existing/planned |
| is_proxy | is_proxy | byte | all | TWN S19 或其他 proxy | existing/planned |
| is_aggregate | is_agg | byte | production/country | aggregate geography/sector | existing/planned |

| Canonical name | Stata name | Type | Grain | 用途 | 状态 |
|---|---|---|---|---|---|
| is_residual | is_resid | byte | country | Mexico residual 等 | existing/planned |
| suppressed_flag | suppressed | byte | production | BEA suppressed | planned M4 |
| confidential_flag | confidentl | byte | production | Mexico confidential | existing/planned |
| allocated_flag | allocated | byte | production | AMNE allocated metric | planned M4 |
| nonadditive_flag | nonadd | byte | all | repeated/rate/stock 不可加总 | planned |

Stata 中所有逻辑变量使用 numeric byte 0/1，并附 value label。不能把 Python boolean 转成字符串 "True"/"False"。

# 3. M2 已验证的政策变量

文件：data/processed/harmonized/product_treatment_hs6_year.parquet。粒度 G_PRODUCT_YEAR。

| Canonical name | Stata name | 定义/单位 | 可加总性 | 用途 | 状态 |
|---|---|---|---|---|---|
| total_us_import_value_2017 | usimp17 | 2017 total U.S. imports，USD | level；映射后可加总 | treatment 分母 | existing validated |
| china_import_value_2017 | chnimp17 | 2017 China imports，USD | level；映射后可加总 | China share 和 weights | existing validated |
| china_import_share_2017 | chnshr17 | chnimp17/usimp17，fraction | nonadditive rate | exposure 分母结构 | existing validated |
| tariff_rate_china_weighted | trf_chw | source HS17 rate 按 China imports 加权 | nonadditive rate | sensitivity/provenance | existing validated |
| tariff_rate_total_import_weighted | trf_usw | source rate 按 total imports 加权 | nonadditive rate | zero-China fallback | existing validated |
| tariff_rate | trf_rate | 最终 statutory rate，fraction | nonadditive rate | policy intensity | existing validated |
| tariff_exposure | trf_expo | sum(China imports × rate)/sum(total imports) | nonadditive rate | 主 gross exposure | existing validated |
| tariffed_product | tariffed | 任一 mapped source code rate > 0 | flag | treatment indicator | existing validated |
| source_hs17_code_count | hs17_n | 汇入 analysis product 的 HS17 数量 | count | audit many-to-one | existing validated |
| policy_source_observed_count | polobs_n | legal panel 中 observed source codes | count | 区分 not-listed 与 missing | existing validated |
| source_tariffed_hs17_count | trfhs17_n | positive-rate source codes | count | treatment composition | existing validated |
| tariff_rate_max_source | trf_max | mapped sources 最大法定 rate | nonadditive rate | denominator=0 fallback | existing validated |
| policy_mapping_status | pol_map | weighted/no denominator/no HS17 weight | category | sample/filter/audit | existing validated |
| historical_exclusions_applied | hist_excl | 当前恒为 0 | flag | 防止误称 realized duties | existing validated |
| four_year_review_2024_included | rev24_in | 当前恒为 0 | flag | 防止误并 2024 review | existing validated |

禁止变量：

| Prohibited name/concept | 原因 |
|---|---|
| realized_tariff_paid from current panel | 当前数据未重建 customs-entry realized duties |
| tariff_exposure_with_exclusions | historical exclusions 尚未应用 |
| 2024_review_hs6 by first-six-digit truncation | modern HTS scope 和 historical HS6 不可直接等同 |

# 4. M2 已验证的贸易金额和 Census 变量

## 4.1 Connector trade

文件：connector_trade_hs6_year.parquet。粒度 G_CONN_PRODUCT_YEAR。

| Canonical name | Stata name | 单位 | 可加总性 | 用途 | 状态 |
|---|---|---|---|---|---|
| china_to_usa_value_usd | cn_us_val | current USD | NONADDITIVE_REPEAT across connector | 共同 China-USA benchmark | existing validated |
| china_to_connector_value_usd | cn_conn_val | current USD | add across products, not connectors unless desired | China input leg | existing validated |
| east_asia_to_connector_value_usd | ea_conn_val | current USD | add across products | East Asia input leg | existing validated |
| connector_to_usa_value_usd | conn_us_val | current USD | add across products/connectors with clear scope | connector export leg | existing validated |
| *_row_count | shortened *_n | count | additive count | 区分 true zero 与 source rows | existing validated |
| east_asia_self_trade_excluded | ea_selfx | flag | flag | KOR/TWN self-trade rule | existing validated |
| baci_absence_treated_as_zero | baci_zrule | flag | flag | complete-grid zero rule | existing validated |
| china_to_usa_common_benchmark | cnus_common | flag | flag | repeated warning | existing validated |
| treatment_repeated_across_connectors | trf_repeat | flag | flag | non-additive warning | existing validated |

## 4.2 Census

月度 grain G_ORIGIN_PRODUCT_MONTH；年度 grain origin_iso3, analysis_hs6, year。

| Canonical name | Stata name | 单位/公式 | 用途 | 状态 |
|---|---|---|---|---|
| import_value_usd | imp_val | current USD | 主进口金额 | existing validated |
| customs_value | cust_val | current USD | customs valuation check | existing validated |
| duties | duties | current USD | collected duties 描述，不等于 Section 301-only duties | existing validated |
| dutiable_value_usd | dutiable | current USD | dutiable base | existing validated |
| observed_month_count | month_n | 0-12 | 年度 cell 的月度覆盖 | existing validated |
| calendar_source_months_complete | cal12_ok | flag | 允许 sparse absence -> zero 的前提 | existing validated |
| quantity_usable | qty_ok | 当前恒为 0 | 禁止 quantity analysis | existing validated |
| is_all_origin_denominator | allorig | 当前恒为 0 | 防止误作全来源分母 | existing validated |

duties/import_value 可以作为描述性 average duty ratio，但不能命名为 Section 301 realized rate，除非另行识别 tariff program 和 entry-level scope。

# 5. M3.1 connector-sector-year 基础聚合变量

状态：EXISTING_VALIDATED。

正式文件：data/processed/harmonized/connector_trade_sector_year.parquet。粒度 G_CONN_SECTOR_YEAR，唯一键为 connector_iso3, analysis_sector, year。

构建证据：

- builder：code/data_processing/build_connector_trade_sector_panel.py；
- 冻结配置：configs/connector_trade_sector_rules.yaml；
- 输入：743,886 行 connector_trade_hs6_year.parquet 与 5,202 行 product_sector_crosswalk.parquet；

- 输出：1,001 行、70 列，13 connectors × 7 sectors × 2014-2024；
- 输出 SHA256：95e547267892c4bc883788542f0e59ba284441b42f4ce1d379aea71a2f978c8c；
- validation：data/validation/connector_trade_sector_panel_validation.json 与 .md，状态 PASS；
- 配置 SHA256：6a313c01d9bfad6e6e6a2cb508bd787f35bdb0eddeb6e4f491a86d61bdd9ee9d；
- 确定性检查：项目 F 盘临时目录第二次构建的 Parquet SHA256 与正式输出完全一致，临时目录已清除；
- Stata 接口：70 个 alias 全部登记、互不重复、最长 14 个字符，均不超过 Stata 的 32 字符限制。本阶段不提前生成 .dta。

## 5.1 金额和覆盖

先用 analysis_hs6 把 M2 panel many-to-one 连接到 canonical product-sector crosswalk。只有 canonical is_analysis_sector_eligible=true 且 sector_mapping_weight=1 的产品，才在 connector_iso3, analysis_sector, year 内求和。M2 内嵌 sector 字段只用于一致性检查，不能取代 canonical crosswalk。

| Canonical name | Stata name | 公式 | 可加总性 | 用途 | 状态 |
|---|---|---|---|---|---|
| china_to_usa_value_usd | cn_us_val | sum_h eligible china_to_usa_value_usd | NONADDITIVE_REPEAT across connector | China decline benchmark | EXISTING_VALIDATED |
| china_to_connector_value_usd | cn_conn_val | sum_h eligible china_to_connector_value_usd | FLOW | China input leg | EXISTING_VALIDATED |
| east_asia_to_connector_value_usd | ea_conn_val | sum_h eligible east_asia_to_connector_value_usd | FLOW | broader source input leg | EXISTING_VALIDATED |
| connector_to_usa_value_usd | conn_us_val | sum_h eligible connector_to_usa_value_usd | FLOW | connector export leg | EXISTING_VALIDATED |
| sector_eligible_product_count | sec_elig_n | sector 内 distinct eligible HS6 | count | sector composition 与 denominator | EXISTING_VALIDATED |

四条流各有独立 diagnostics；<flow> 分别是 china_to_connector、east_asia_to_connector、connector_to_usa 和 china_to_usa：

| Canonical pattern | Stata aliases | 公式和含义 | 状态 |
|---|---|---|---|
| <flow>_value_usd_row_count | cn_conn_src_n, ea_conn_src_n, conn_us_src_n, cn_us_src_n | eligible products 的 M2 source-row count 之和 | EXISTING_VALIDATED |
| <flow>_positive_product_count | cn_conn_pos_n, ea_conn_pos_n, conn_us_pos_n, cn_us_pos_n | sector 内 <flow>_value_usd > 0 的 eligible HS6 数 | EXISTING_VALIDATED |
| <flow>_zero_source_product_count | cn_conn_zsrc_n, ea_conn_zsrc_n, conn_us_zsrc_n, cn_us_zsrc_n | sector 内 M2 <flow>_row_count = 0 的 eligible HS6 数；保留 source absence，不等同 observed zero | EXISTING_VALIDATED |

excluded、outside-manufacturing 和 residual products 没有可用于 sector sums 的有效 sector weight，因此 coverage 在 connector_iso3, year 粒度计算后跨 7 个 sector 重复。所有 coverage 字段都带 coverage_repeated_across_sectors=true，不能跨 sector 求和。

| Canonical name | Stata name | 公式 | 可加总性 | 状态 |
|---|---|---|---|---|
| excluded_product_count | excl_prod_n | connector-year 内 965 个非 eligible 产品的 distinct count | NONADDITIVE_REPEAT across sector | EXISTING_VALIDATED |
| china_to_connector_excluded_value_usd | cn_conn_excl | 非 eligible 产品的 China→connector value sum | NONADDITIVE_REPEAT across sector | EXISTING_VALIDATED |
| china_to_connector_included_value_share | cn_conn_incl_sh | eligible / full-universe China→connector value | NONADDITIVE_RATE | EXISTING_VALIDATED |
| east_asia_to_connector_excluded_value_usd | ea_conn_excl | 非 eligible 产品的 East Asia→connector value sum | NONADDITIVE_REPEAT across sector | EXISTING_VALIDATED |
| east_asia_to_connector_included_value_share | ea_conn_incl_sh | eligible / full-universe East Asia→connector value | NONADDITIVE_RATE | EXISTING_VALIDATED |
| connector_to_usa_excluded_value_usd | conn_us_excl | 非 eligible 产品的 connector→USA value sum | NONADDITIVE_REPEAT across sector | EXISTING_VALIDATED |
| connector_to_usa_included_value_share | conn_us_incl_sh | eligible / full-universe connector→USA value | NONADDITIVE_RATE | EXISTING_VALIDATED |

| Canonical name | Stata name | 公式 | 可加总性 | 状态 |
|---|---|---|---|---|
| china_to_usa_excluded_value_usd | cn_us_excl | 非 eligible 产品的 China→USA value sum；同时跨 connector 重复 | NONADDITIVE_REPEAT | EXISTING_VALIDATED |
| china_to_usa_included_value_share | cn_us_inclsh | eligible / full-universe China→USA value；同时跨 connector 重复 | NONADDITIVE_RATE | EXISTING_VALIDATED |

## 5.2 sector-level treatment

先对 analysis_hs6, year 去除 connector 复制，再在 sector-year 重算分子和分母。Policy rate/exposure 不能按产品 rate 直接求和。

| Canonical name | Stata name | 公式 | 用途 | 状态 |
|---|---|---|---|---|
| sector_total_us_import_value_2017 | sec_usimp17 | sum_h usimp17_h | transparent denominator | EXISTING_VALIDATED |
| sector_china_import_value_2017 | sec_cnimp17 | sum_h chnimp17_h | China weight denominator | EXISTING_VALIDATED |
| sector_tariff_exposure_numerator_usd | sec_trf_num | sum_h(chnimp17_h × rate_h,t) | common transparent numerator | EXISTING_VALIDATED |
| sector_tariff_rate_china_weighted | sec_trf_cw | sec_trf_num / sec_cnimp17 | 法定 rate intensity | EXISTING_VALIDATED |
| sector_tariff_exposure | sec_expo | sec_trf_num / sec_usimp17 | 主 gross policy exposure | EXISTING_VALIDATED |
| sector_china_import_share_2017 | sec_cnshr17 | sec_cnimp17 / sec_usimp17 | baseline China dependence | EXISTING_VALIDATED |
| sector_treatment_product_count | sec_trt_n | distinct eligible HS6 | policy count denominator | EXISTING_VALIDATED |
| policy_denominator_observed_product_count | pol_denobs_n | usimp17_h > 0 且非缺失的 eligible HS6 数 | denominator audit | EXISTING_VALIDATED |
| tariffed_product_count | trf_prod_n | tariffed_product=true 的 eligible HS6 数 | breadth numerator | EXISTING_VALIDATED |
| tariffed_china_import_value_2017 | trf_cnimp17 | tariffed eligible HS6 的 China imports sum | China treatment mass | EXISTING_VALIDATED |
| tariffed_total_us_import_value_2017 | trf_usimp17 | tariffed eligible HS6 的 total U.S. imports sum | total-import treatment mass | EXISTING_VALIDATED |
| tariffed_product_share_count | trf_psh_n | trf_prod_n / sec_trt_n | breadth | EXISTING_VALIDATED |
| tariffed_product_share_china_value | trf_psh_c | trf_cnimp17 / sec_cnimp17 | China treatment mass share | EXISTING_VALIDATED |
| tariffed_product_share_us_value | trf_psh_u | trf_usimp17 / sec_usimp17 | total-import treatment mass share | EXISTING_VALIDATED |

缺少 denominator 或 HS2017 mapping 的产品不静默转成普通零 denominator：

| Canonical name | Stata name | 公式/含义 | 状态 |
|---|---|---|---|
| treatment_missing_denominator_product_count | trt_missden_n | usimp17_h 缺失或不大于 0 的 eligible HS6 数 | EXISTING_VALIDATED |
| treatment_zero_denominator_product_count | trt_zeroden_n | policy_mapping_status=mapped_hs17_no_2017_import_denominator 的 HS6 数；全样本 17 个 distinct products | EXISTING_VALIDATED |
| treatment_no_hs17_mapping_product_count | trt_nohs17_n | policy_mapping_status=no_hs17_baci_weight_mapping 的 HS6 数；全样本 11 个 distinct products | EXISTING_VALIDATED |
| treatment_missing_denominator_product_share | trt_miss_sh | missing denominator count / sec_trt_n | EXISTING_VALIDATED |
| treatment_zero_denominator_product_share | trt_zero_sh | zero denominator count / sec_trt_n | EXISTING_VALIDATED |
| treatment_no_hs17_mapping_product_share | trt_nohs17_sh | no-HS2017-mapping count / sec_trt_n | EXISTING_VALIDATED |

主回归/排名应优先使用 sector_tariff_exposure，同时把 weighted rate 和 tariffed shares 作为 robustness，不把它们互相替代。

## 5.3 标签、版本和不可加总 flags

| Canonical name | Stata name | 内容/规则 | 状态 |
|---|---|---|---|
| connector_group | conn_group | core/comparator 分组 | EXISTING_VALIDATED |
| connector_baci_code | conn_baci | BACI reporter/partner code | EXISTING_VALIDATED |
| connector_is_proxy | is_proxy | TWN 使用 BACI S19 proxy；TWN 的 77 行全部为 1 | EXISTING_VALIDATED |
| analysis_sector_status | sec_status | canonical sector status | EXISTING_VALIDATED |
| trade_value_unit | val_unit | current USD | EXISTING_VALIDATED |
| treatment_concept | trt_concept | gross statutory initial actions | EXISTING_VALIDATED |
| historical_exclusions_applied | hist_excl | 恒为 0 | EXISTING_VALIDATED |
| four_year_review_2024_included | rev24_in | 恒为 0 | EXISTING_VALIDATED |
| review_2024_treatment | rev24_trt | separate exact-HTS extension | EXISTING_VALIDATED |
| treatment_version | trt_ver | M2 treatment version | EXISTING_VALIDATED |
| is_analysis_sector_eligible | ana_ok | 本输出恒为 1 | EXISTING_VALIDATED |
| sector_mapping_weight | sec_wt | 本输出恒为 1 | EXISTING_VALIDATED |
| baci_absence_treated_as_zero | baci_zrule | complete-grid BACI absence rule | EXISTING_VALIDATED |
| quantity_included | qty_in | 恒为 0；本输出不含 quantity | EXISTING_VALIDATED |
| china_to_usa_common_benchmark | cnus_common | China→USA benchmark 跨 connector 重复 | EXISTING_VALIDATED |
| treatment_repeated_across_connectors | trt_repeat | sector treatment 跨 connector 重复 | EXISTING_VALIDATED |
| coverage_repeated_across_sectors | cov_repeat | connector-year coverage 跨 sector 重复 | EXISTING_VALIDATED |
| common_benchmarks_not_additive_across_connectors | bm_nonadd | 禁止跨 connector 求和共同 benchmark | EXISTING_VALIDATED |
| product_mapping_version | prod_mapver | canonical product-sector mapping version | EXISTING_VALIDATED |
| m3_sector_aggregation_version | m3_aggver | connector_trade_sector_rules_v1 | EXISTING_VALIDATED |
| schema_version | schema_ver | connector_trade_sector_year_v1 | EXISTING_VALIDATED |

## 5.4 M3.1 验证结论和解释边界

- 两个输入 SHA256 与冻结配置一致；输入、crosswalk 和输出主键重复均为 0；全部 HS6 成功匹配，M2 内嵌 sector 字段与 canonical crosswalk 不一致数为 0。
- eligible HS6 数固定为 autos 64、electrical equipment 177、electronics 252、machinery 467、medical devices 32、other manufacturing 2,422、textiles/apparel 823。
- 四条 flow 的 product-level eligible totals 与 sector totals 差额均为 USD 0；eligible + excluded = full universe 的 connector-year 最大差额均为 USD 0。
- 九个 policy rate/share 字段独立公式复算最大绝对误差均为 0，小于 1e-12。
- China→USA benchmark、policy 和 coverage 的重复一致性不一致数均为 0。
- 当前输出只含年度 level、coverage 和 sector treatment。它不含 pre/shock/post window、growth、small-base、replacement ratio、ranking、production-side 或 structural moments。

# 6. M3.2 时间窗口变量

状态：EXISTING_VALIDATED。

正式文件：

1. data/processed/harmonized/connector_trade_sector_window.parquet：273 行、79 列，粒度 G_CONN_SECTOR_WINDOW，唯一键 connector_iso3, analysis_sector, window_id；
2. data/processed/harmonized/connector_trade_sector_window_comparison.parquet：91 行、142 列，粒度 G_CONN_SECTOR_COMPARISON，唯一键 connector_iso3, analysis_sector。

构建证据：

- builder：code/data_processing/build_connector_trade_sector_window_panel.py；
- 冻结配置：configs/connector_trade_sector_window_rules.yaml，SHA256 95d42ee146b1e0cf6e98261b46e5db52bf81000770d12d55da3371350d50058b；
- 冻结输入：1,001 × 70 的 M3.1 年度 panel，SHA256 95e547267892c4bc883788542f0e59ba284441b42f4ce1d379aea71a2f978c8c；
- 时间定义：configs/time_windows.yaml，SHA256 d1daeed066db7cf734c15c804f31eaaf66a7c07b335f3c227a6c9effffd6ebb5；
- long 输出 SHA256：310da7f289d8760d3ead34215a294be18275c758a2084cfd9f905988b695e704；
- comparison 输出 SHA256：be5f3a025a81d36d5a9bb1bd137d4bed9325a68ab5f1cb7f4d946a2373fbeb04；
- validation：data/validation/connector_trade_sector_window_validation.json 与 .md，状态 PASS；
- 确定性检查：项目 F 盘第二次构建的两张 Parquet 分别与正式候选逐字节 hash 相同，候选和 determinism 临时目录均已清除；
- Stata 接口：long 79 个、comparison 142 个 resolved aliases 全部完整、互不重复，最长 13 字符。本阶段仍不生成 .dta。

## 6.1 窗口定义和 long annual-mean levels

不同窗口长度不相等：pre 4 年、shock 2 年、post 5 年。因此 level comparison 使用完整年度观察的算术平均值，不使用窗口总和。

| window_id | 年份 | window_order | expected/observed years | 状态 |
|---|---|---|---|---|
| pre | 2014-2017 | 1 | 4 / 4 | window_complete=true |
| shock | 2018-2019 | 2 | 2 / 2 | window_complete=true |
| post | 2020-2024 | 3 | 5 / 5 | window_complete=true |

所有 91 个 connector-sector cells 在三个窗口都完整，没有 partial-window mean。Long 表的窗口字段为：

| Canonical name | Stata name | 定义 | 状态 |
|---|---|---|---|
| window_id | window_id | pre/shock/post | EXISTING_VALIDATED |
| window_order | win_order | 1/2/3 | EXISTING_VALIDATED |
| window_start_year / window_end_year | win_start / win_end | 冻结端点 | EXISTING_VALIDATED |
| expected_year_count / observed_year_count | exp_yn / obs_yn | 预期与实际 distinct years | EXISTING_VALIDATED |
| window_complete | win_complete | observed=expected | EXISTING_VALIDATED |
| window_level_is_annual_mean | win_mean | 本输出恒为 1 | EXISTING_VALIDATED |

四条流在 long 表使用下列 family。<flow>/<prefix> 对应 china_to_connector/cncn、east_asia_to_connector/eacn、connector_to_usa/cnusx、china_to_usa/cnus。

| Canonical pattern | Stata pattern | 公式/含义 | 状态 |
|---|---|---|---|
| <flow>_value_usd_annual_mean | <prefix>_val | 完整窗口年度金额算术平均 | EXISTING_VALIDATED |
| <flow>_source_row_count_annual_mean | <prefix>_src_n | M3.1 flow-specific source-row counts 的年度平均 | EXISTING_VALIDATED |
| <flow>_positive_product_count_annual_mean | <prefix>_pos_n | positive eligible HS6 count 的年度平均 | EXISTING_VALIDATED |
| <flow>_zero_source_product_count_annual_mean | <prefix>_zsrc_n | source-row count=0 eligible HS6 count 的年度平均；不等同 observed zero | EXISTING_VALIDATED |

年度 count 的窗口平均可能为非整数，因此 long 表中这些 annual-mean diagnostics 使用 double；底层年度 count 没有被修改。

## 6.2 Coverage 使用 ratio of sums

非 eligible 产品没有 sector weight。Coverage 仍先在 connector_iso3, year 计算，再聚合到 connector_iso3, window_id 并跨 sector 重复；coverage_repeated_across_sectors=true 继续禁止跨 sector 加总。

| Canonical pattern | Stata pattern | 公式 | 可加总性 | 状态 |
|---|---|---|---|---|
| excluded_product_count | excl_prod_n | 窗口内固定的非 eligible 产品数 965 | NONADDITIVE_REPEAT | EXISTING_VALIDATED |
| <flow>_excluded_value_usd_annual_mean | <prefix>_excl | 非 eligible 年度金额的窗口平均 | NONADDITIVE_REPEAT across sector | EXISTING_VALIDATED |
| <flow>_included_value_share | <prefix>_inclsh | sum_t eligible_value_t / sum_t(eligible_value_t + excluded_value_t) | NONADDITIVE_RATE | EXISTING_VALIDATED |
| included_value_share_uses_ratio_of_sums | incl_rsum | 本输出恒为 1，禁止误解为 annual shares 的简单平均 | FLAG | EXISTING_VALIDATED |

## 6.3 Window policy summaries

Policy denominator 仍是透明的 2017 sector denominator。时间变化的 numerator/count/value 先取完整窗口年度平均，再重算 rate/share；不平均年度 rate/share。

| Canonical name | Stata name | 公式 | 状态 |
|---|---|---|---|
| sector_tariff_exposure_numerator_usd_annual_mean | sec_trf_num | mean_t sector_tariff_exposure_numerator_usd_t | EXISTING_VALIDATED |
| sector_total_us_import_value_2017 | sec_usimp17 | 固定 2017 total-import denominator | EXISTING_VALIDATED |
| sector_china_import_value_2017 | sec_cnimp17 | 固定 2017 China-import denominator | EXISTING_VALIDATED |
| sector_tariff_rate_china_weighted | sec_trf_cw | annual-mean numerator / sec_cnimp17 | EXISTING_VALIDATED |
| sector_tariff_exposure | sec_expo | annual-mean numerator / sec_usimp17 | EXISTING_VALIDATED |
| sector_china_import_share_2017 | sec_cnshr17 | sec_cnimp17 / sec_usimp17 | EXISTING_VALIDATED |
| tariffed_product_count_annual_mean | trf_prod_n | 年度 tariffed eligible-product count 的窗口平均 | EXISTING_VALIDATED |
| tariffed_china_import_value_2017_annual_mean | trf_cnimp17 | tariffed China-import mass 的年度平均 | EXISTING_VALIDATED |
| tariffed_total_us_import_value_2017_annual_mean | trf_usimp17 | tariffed total-import mass 的年度平均 | EXISTING_VALIDATED |
| tariffed_product_share_count | trf_psh_n | annual-mean tariffed count / treatment count | EXISTING_VALIDATED |
| tariffed_product_share_china_value | trf_psh_c | annual-mean tariffed China value / sec_cnimp17 | EXISTING_VALIDATED |
| tariffed_product_share_us_value | trf_psh_u | annual-mean tariffed total value / sec_usimp17 | EXISTING_VALIDATED |

Treatment denominator observed/missing/zero/no-HS2017 counts和 shares 保留为窗口字段。每个窗口跨 sector 加总仍为 17 个 zero-denominator eligible products 和 11 个 no-HS2017-mapping eligible products。Pre 的 91 个 connector-sector policy exposure 全部为 0；本模块没有对 policy exposure 构造 change、percentage growth 或 log growth。

## 6.4 Comparison levels、changes 和 growth

对任意非负年度 flow x_t:

```
x_pre  = mean(x_t, 2014-2017)
x_shk  = mean(x_t, 2018-2019)
x_post = mean(x_t, 2020-2024)

d_x_post_pre = x_post - x_pre
g_dh_x_post_pre = 2 * (x_post - x_pre) / (x_post + x_pre)
g_log_x_post_pre = ln(x_post) - ln(x_pre), only if both > 0
```

g_dh 是推荐 primary growth：无量纲、允许一端为 0、范围 [-2,2]。g_log 只用于 positive-base robustness。禁止把 undefined log growth 自动设为 0。

Comparison 表对每条贸易腿构造：

| Family | Canonical stem | Stata prefix | Pre/shock/post | Change | DH | Log |
|---|---|---|---|---|---|---|
| China -> USA | china_to_usa | cnus | cnus_pre/shk/post | d_cnus | gdh_cnus | gln_cnus |
| China -> connector | china_to_connector | cncn | cncn_pre/shk/post | d_cncn | gdh_cncn | gln_cncn |
| East Asia -> connector | east_asia_to_connector | eacn | eacn_pre/shk/post | d_eacn | gdh_eacn | gln_eacn |
| connector -> USA | connector_to_usa | cnusx | cnusx_pre/shk/post | d_cnusx | gdh_cnusx | gln_cnusx |

完整 canonical patterns 是 <flow>_value_usd_pre/shock/post、<flow>_value_usd_change_post_pre、<flow>_value_usd_dh_growth_post_pre 和 <flow>_value_usd_log_growth_post_pre。Comparison 表还展开四条流的 excluded annual mean、included-value share，以及九个随窗口变化的 policy summaries；固定 2017 denominators 和 denominator-exception counts/shares只保留一份，避免无意义的 pre/shock/post 复制。

## 6.5 Zero/defined flags 和 M3.3 边界

| Canonical pattern | Stata pattern | 定义 | 状态 |
|---|---|---|---|
| <flow>_pre_zero_flag | <prefix>_pre0 | pre annual mean = 0 | EXISTING_VALIDATED |
| <flow>_post_zero_flag | <prefix>_post0 | post annual mean = 0 | EXISTING_VALIDATED |
| <flow>_dh_growth_defined | <prefix>_dhok | pre+post != 0 | EXISTING_VALIDATED |
| <flow>_log_growth_defined | <prefix>_logok | pre>0 and post>0 | EXISTING_VALIDATED |
| <flow>_years_observed_pre/shock/post | <prefix>_preyn/shkyn/postyn | 实际 distinct years | EXISTING_VALIDATED |

边界规则已实现：

- 两端都为 0：DH 和 log 均缺失；
- 只有一端为 0：DH 为 ±2，log 缺失；
- 两端都为正：DH 和 log 均定义；
- 负 level change 和负 growth 原样保留，不截断。

当前数据四条流的 pre-zero 和 post-zero 数都为 0，每条流 91 个 log growth 全部定义。负 post-pre change 数分别为 China→connector 3、East Asia→connector 5、connector→USA 6、China→USA 26。DH 全部位于 [-2,2]，没有 infinite。

small_base_flag 仍是 PLANNED_M3，M3.2 不设阈值、不标异常、不 winsorize。M3.3 必须另行冻结 absolute USD threshold、sector-relative threshold 和 no-threshold/DH-only robustness。推荐主排名继续同时使用 level change 和 DH growth，不依赖单一百分比增长。

## 6.6 M3.2 validation 和解释边界

- M3.1 input hash、70-column Arrow schema signature、1,001 行和年度主键全部匹配冻结配置；
- 每个年度观察恰好属于一个窗口，source window definition 与 frozen definition 完全一致；
- long/comparison 主键重复均为 0，行数严格为 273/91；
- 四条流 annual mean × years 与 M3.1 annual sum 的最大金额差额均为 USD 0；source-row、positive-product 和 zero-source diagnostics 的 count reconciliation gap 均为 0；
- comparison 的 pre/shock/post levels 与 long 完全一致；change、DH 和 log 独立复算最大绝对误差均为 0；
- coverage ratio-of-sums 和所有 policy numerator/denominator/rate/share 独立复算误差均为 0；
- China→USA/policy 跨 connector、coverage 跨 sector 的 inconsistency count 均为 0；
- TWN 有 21 个 long rows、7 个 comparison rows，proxy false count 为 0；
- long/comparison required numeric missing count 均为 0；
- 两张 Parquet 的第二次确定性构建 SHA256 分别完全一致，candidate 和 determinism 临时目录已清除。

M3.2 可以作为 M3.3 的唯一输入接口之一。它不能用于声称 small-base 处理、异常增长决策、replacement ratio、connector ranking、production relocation 或 structural moments 已完成。

# 7. M3.3-M3.6 贸易重组和排名输入变量

## 7.1 Import replacement

```
china_us_decline = max(cnus_pre - cnus_post, 0)
connector_us_gain = cnusx_post - cnusx_pre
replacement_ratio = connector_us_gain / china_us_decline
```

replacement_ratio 只有在 china_us_decline 大于明确的 denominator threshold 时定义；否则 missing。主结果同时保留 level gain 和 decline，不能只报告 ratio。

| Canonical name | Stata name | 用途 | 状态 |
|---|---|---|---|
| china_us_decline_usd | cnus_drop | China displacement level | planned M3 |
| connector_us_gain_usd | conn_gain | connector gain level | planned M3 |
| replacement_ratio | repl_ratio | connector gain 相对 China decline | planned M3 |
| replacement_ratio_defined | repl_ok | denominator 是否可用 | planned M3 |
| replacement_capped_for_rank | repl_cap | 仅 ranking 的 winsorized/capped version | planned M3 |

## 7.2 Source-to-connector input growth

| Canonical name | Stata name | 定义 |
|---|---|---|
| china_input_gain_usd | cnin_gain | China -> connector post-pre level change |
| east_asia_input_gain_usd | eain_gain | East Asia -> connector post-pre level change |
| china_input_dh_growth | gdh_cnin | China input DH growth |
| east_asia_input_dh_growth | gdh_eain | East Asia input DH growth |
| source_export_ratio_pre | srcx_pre | source-to-connector / connector-to-USA，pre |
| source_export_ratio_post | srcx_post | 同一 ratio，post |
| source_export_ratio_change | d_srcx | post-pre |

ratio denominator 为 0 时保持 missing；不能用 arbitrary epsilon 生成正式值。若 ranking 需要 finite score，先用 percentile of level/growth components，而不是偷偷给 ratio 加常数。

## 7.3 Product-level co-movement

在 connector-sector 内先对每个 HS6 计算 pre/post changes，再聚合：

| Canonical name | Stata name | 公式/用途 |
|---|---|---|
| both_legs_positive_product_share | both_psh | China/East Asia input gain >0 且 connector-US gain >0 的 HS6 share |
| both_legs_positive_value_share | both_vsh | 上述产品按 pre/post trade value 加权 share |
| china_decline_connector_gain_share | rerout_psh | China-US decline 且 connector-US gain 的 product share |
| triple_condition_product_share | triple_psh | China-US decline、source input gain、connector-US gain 同时成立 |
| product_change_correlation | prod_corr | source-input change 与 connector-export change 的 HS6 correlation |
| weighted_product_change_correlation | wprod_cor | 以 pre China-US 或 connector trade weights 加权 |

相关系数至少要求有效 products 达到 config 中的 minimum count；否则 missing + corr_low_n_flag=1。

## 7.4 Exposure overlap

| Canonical name | Stata name | 公式 |
|---|---|---|
| connector_export_tariff_overlap_pre | ovlp_pre | pre connector-US exports 中 tariffed/exposed product 的 value share |
| connector_export_tariff_overlap_post | ovlp_post | post share |
| connector_export_exposure_weighted | ovlp_expo | connector export weights 下的 product tariff exposure |
| gain_in_high_exposure_products_share | gain_hiexp | export gain 来自 high-exposure products 的份额 |

“high exposure” threshold 必须在 config 中固定，例如 top quartile 或 positive exposure；两种定义均应报告。

## 7.5 稳定性和规模

| Canonical name | Stata name | 用途 |
|---|---|---|
| pre_trade_scale_usd | scale_pre | 排名时防止只由小基数增长驱动 |
| post_trade_scale_usd | scale_post | 经济规模 |
| annual_growth_sd | growth_sd | 年度增长波动 |
| positive_growth_year_share | posyr_sh | post 期间正增长年份比例 |
| rank_window_sensitivity | rank_sens | 替代窗口下 rank 变化 |
| source_bloc_sensitivity | bloc_sens | China-only 与 East-Asia source 定义差异 |
| exposure_definition_sensitivity | expo_sens | rate/exposure/tariffed share 差异 |

# 8. M4 production-side 标准化变量

目标文件：production_validation_long.parquet。采用长表保存来源、指标与单位；禁止把不同概念合并相加。

## 8.1 通用 long schema

| Canonical name | Stata name | 定义 |
|---|---|---|
| country_iso3 | country_iso3 | host/country |
| analysis_sector | ana_sector | 共同 sector |
| year | year | 年份 |
| source_name | source | icio/amne/bea_mne/bea_fdi/mexico_fdi |
| metric_name | metric | 指标名 |
| value | value | 原单位数值 |
| unit | unit | current USD/current million USD/thousand persons/fraction |
| frequency | freq | annual/quarterly/H1 |
| direction | direction | FDI inward/outward 或 affiliate trade direction |
| parent_country_iso3 | parent_iso3 | AMNE parent，适用时 |
| direct_or_allocated | direct_alloc | direct/allocated/unavailable |
| suppressed_flag | suppressed | BEA |
| confidential_flag | confidentl | Mexico |
| aggregate_flag | is_agg | geography/sector aggregate |
| proxy_flag | is_proxy | coarse source-sector proxy |
| source_file | src_file | provenance |

## 8.2 ICIO metrics

| Metric | Stata metric label | Unit | Additivity/use |
|---|---|---|---|
| gross_output | gross_output | current million USD | production level |
| value_added | value_added | current million USD | production validation；允许 accounting negatives |
| exports | exports | current million USD | source-consistent export level |
| imports | imports | current million USD | source-consistent import level |
| intermediate_inputs | intermed_inputs | current million USD | IO intensity |
| taxes_less_subsidies | tax_less_sub | current million USD | accounting；允许负值 |
| accounting_residual | acct_resid | current million USD | quality diagnostic |

ICIO 只有 2017、2018、2022。禁止线性插值得到 2019 或 annual panel。

## 8.3 AMNE metrics

| Metric | Unit | Direct/allocated | 用途 |
|---|---|---|---|
| affiliate_output | current million USD | direct bilateral | MP activity 主指标 |
| affiliate_value_added | current million USD | allocated to parent | production validation |
| affiliate_exports | current million USD | allocated | affiliate trade |
| affiliate_imports | current million USD | allocated | affiliate input |
| foreign_affiliate_output_total | current million USD | host-sector total | foreign activity scale |
| parent_affiliate_output_share | fraction | derived direct output share | allocation provenance |
| affiliate_employment | unavailable | unavailable | PROHIBITED as zero/imputed |

## 8.4 BEA MNE metrics

| Metric | Unit | 用途 |
|---|---|---|
| employment | thousand employees | labor validation |
| sales | million current USD | affiliate activity |
| value_added | million current USD | production |
| capital_expenditures | million current USD | investment/activity |
| goods_exports_to_us | million current USD | affiliate -> U.S. direction |
| goods_imports_from_us | million current USD | U.S. -> affiliate direction |

每个 metric 同时保留 suppression 和 small-value flag。aggregate geography 先排除。Transportation equipment 若作为 autos robustness，必须 proxy_flag=1。

## 8.5 BEA FDI metrics

| Metric | Temporal type | Unit | 规则 |
|---|---|---|---|
| position | stock | million current USD | 不跨年求和 |
| flows | annual flow | million current USD | 允许负 reversal |
| income | annual flow | million current USD | 与 position/flow 分开 |

必须保留 direction。不能构造 total_fdi = position + flows + income。

## 8.6 Mexico FDI metrics

| Metric | Stata name | 规则 |
|---|---|---|
| fdi_quarterly_usd | fdi_q_usd | quarterly flow，负值保留 |
| fdi_annual_usd | fdi_a_usd | 仅四季度完整且均非 confidential 时求和 |
| fdi_h1_usd | fdi_h1_usd | 2024 与历史 H1-to-H1 比较 |
| fdi_complete_year_flag | fdi_yr_ok | 2006-2023 full-year；2024 detail false |
| fdi_confidential_quarter_count | fdi_conf_n | confidential quarter count |
| fdi_disclosed_composition_share | fdi_discsh | 只描述 disclosed cells，不能解释为完整 national share |

Mexico national total 与 detailed composition 必须存为不同 source/metric，不能用 disclosed detail 替代全国 total。

# 9. M4-M5 production changes 和覆盖变量

对每个 source-metric 在其实际可用年份内构造变化，不能强迫所有来源使用同一连续窗口。

| Canonical name | Stata name | 公式/规则 |
|---|---|---|
| value_level | value | 原始 level |
| value_lag | value_lag | 同 source-country-sector-metric 上一期 |

| Canonical name | Stata name | 公式/规则 |
|---|---|---|
| value_change | d_value | value-value_lag |
| value_dh_growth | gdh_value | 非负 level 适用；signed FDI 不使用 DH level formula |
| value_log_growth | gln_value | 两期均正时 |
| signed_flow_change | d_flow | signed FDI flow 的 level change |
| post_pre_mean_change | d_postpre | 在 source 可覆盖窗口内年度平均差 |
| coverage_year_count | cover_yn | 有效年份数 |
| missing_value_count | miss_n | missing 数 |
| suppressed_value_count | supp_n | suppressed 数 |
| confidential_value_count | conf_n | confidential 数 |

M5 coverage flags：

| Canonical name | Stata name |
|---|---|
| has_trade | has_trade |
| has_icio | has_icio |
| has_amne | has_amne |
| has_bea_mne | has_beamne |
| has_bea_fdi | has_beafdi |
| has_mexico_fdi | has_mexfdi |
| has_any_production | has_prod |
| has_direct_activity_measure | has_direct |
| has_only_financial_evidence | fin_only |
| source_count_available | source_n |

has_only_financial_evidence=1 的 cell 不能标为 production confirmed。

# 10. M6 connector ranking

## 10.1 组件

推荐先把 raw components 转为同一比较组内的 percentile rank [0,1]，减少单位和极端值影响，同时永久保留 raw variables。

| Component | Candidate raw inputs | 经济含义 |
|---|---|---|
| score_exposure | exposure overlap、sector tariff exposure | 是否集中在受冲击产品 |
| score_us_gain | connector-US level gain、DH growth | 对美出口扩张 |
| score_source_gain | China/East-Asia input gain、DH growth | source-to-connector link |
| score_replacement | China decline + connector gain、replacement ratio | U.S. import replacement |
| score_comovement | both/triple product share、weighted correlation | 两条贸易腿同步性 |
| score_scale | pre/post trade scale | 经济规模 |
| score_stability | positive-year share、negative volatility | 时间稳定性 |
| score_production | direct activity measures | 生产侧验证 |

## 10.2 至少两种排名方案

方案 A，推荐主结果：透明 equal-weight percentile score。

```
rank_score_equal =
mean(score_exposure,
    score_us_gain,
    score_source_gain,
    score_replacement,
```

```
    score_comovement,
    score_scale,
    score_stability)
```

优点：透明、可复核、对单位不敏感。缺点：默认各维度同等重要。

方案 B，推荐稳健性：rerouting-core score，减少规模权重。

```
rank_score_rerouting =
0.20 exposure
+ 0.20 us_gain
+ 0.20 source_gain
+ 0.20 comovement
+ 0.10 replacement
+ 0.05 scale
+ 0.05 stability
```

优点：更贴近两段式 rerouting 机制。缺点：权重含研究判断，必须报告替代权重。

production evidence 不建议直接混入主 trade ranking；更清楚的做法是先得到 trade ranking，再给出 production_validation_grade。如果合成总分，必须作为第三套扩展结果。

## 10.3 Ranking outputs

| Canonical name | Stata name | 状态 |
|---|---|---|
| rank_score_equal | score_eq | planned M6 |
| rank_position_equal | rank_eq | planned M6 |
| rank_score_rerouting | score_rr | planned M6 |
| rank_position_rerouting | rank_rr | planned M6 |
| rank_top3_equal | top3_eq | planned M6 |
| rank_top3_rerouting | top3_rr | planned M6 |
| rank_stability_across_specs | rank_stab | planned M6 |
| rank_component_min | score_min | planned robustness |
| ranking_spec | rank_spec | planned M6 |

所有 rank 必须在包含 core connectors 和 comparators 的完整共同样本上计算，不能先删掉低排名观察。

# 11. M6 rerouting facts 和 production validation

## 11.1 Rerouting classification

| Canonical name | Stata name | 定义 |
|---|---|---|
| china_us_decline_flag | cnus_dropf | China-US post < pre，且 level decline 达阈值 |
| connector_us_gain_flag | conn_gainf | connector-US post > pre |
| source_input_gain_flag | src_gainf | China/East Asia -> connector post > pre |
| rerouting_two_leg_flag | rr_2leg | connector gain + source input gain |
| rerouting_three_condition_flag | rr_3cond | China decline + connector gain + source input gain |
| high_exposure_rerouting_flag | rr_hiexp | triple condition + high exposure |
| rerouting_evidence_grade | rr_grade | none/weak/moderate/strong，规则写入 config |

Evidence grade 必须由预先固定的规则生成，不能根据结果手工改标签。

## 11.2 Production validation

| Canonical name | Stata name | 定义 |
|---|---|---|
| output_growth_positive | out_pos | ICIO/direct output change >0 |
| value_added_growth_positive | va_pos | VA change >0 |
| affiliate_activity_growth_positive | aff_pos | AMNE/BEA activity >0 |
| employment_growth_positive | emp_pos | available employment >0 |
| capex_growth_positive | capex_pos | BEA capex >0 |
| fdi_flow_increase | fdi_pos | signed flow change >0 |
| direct_activity_signal_count | direct_n | output/VA/affiliate/employment/capex positive count |
| financial_signal_count | fin_n | FDI flow/position/income signals |
| production_confirmed_flag | prod_conf | 至少一个 direct activity signal，不能只靠 FDI |
| financial_support_only_flag | fin_supp | 只有 financial evidence |
| production_validation_grade | prod_grade | none/financial-only/partial/multiple-source |

推荐 grade：

- none：无生产或金融支持；
- financial_only：只有 FDI；
- partial_direct：至少一个 direct activity metric 同向；
- multiple_direct：至少两个不同 source 或不同 direct activity families 同向；
- conflicting：trade 与 production signals 方向冲突。

# 12. Structural moments 和 Scenario A/B/C

必须区分“观测 moment”和“model shock/parameter”。数据不会直接观测 hgamma。

## 12.1 Trade-cost candidates

| Canonical name | Stata name | 公式 | 状态 |
|---|---|---|---|
| gross_tariff_factor_hat | htau_gross | (1+tariff_rate_post)/(1+tariff_rate_pre) | structural target only |
| exposure_factor_hat | htau_expo | (1+exposure_post)/(1+exposure_pre) | robustness candidate |
| tariff_rate_change_pp | d_trf_pp | 100*(rate_post-rate_pre) | descriptive moment |
| tariff_exposure_change_pp | d_exp_pp | 100*(expo_post-expo_pre) | descriptive moment |

最终 h_tau 选择必须与模型中 iceberg trade cost 或 tariff wedge 的定义一致，不能仅因为变量名相似就直接导入。

## 12.2 MP/production targets

| Moment | 观测来源 | 用途 |
|---|---|---|
| foreign affiliate output share change | AMNE | incumbent MP-intensive adjustment |
| affiliate output growth | AMNE | Scenario B target |
| affiliate sales/VA/employment/capex growth | BEA MNE | direct activity validation |
| gross output/value added growth | ICIO | broader production response |
| FDI position/flow change | BEA/Mexico | Scenario C financial upper-bound support |

Candidate names：

- mp_affiliate_output_share_change -> d_affshr
- mp_affiliate_output_dh_growth -> gdh_affout
- bea_affiliate_activity_index -> bea_affidx
- production_activity_index -> prod_idx

- fdi_financial_index -> fdi_idx

这些是 moments，不是 hgamma 本身。模型校准后才生成：

- hgamma_scenario_b
- hgamma_scenario_c

并必须记录 calibration mapping、normalization、elasticity、target source 和 solver result。

## 12.3 Structural moment table

目标字段：

| Field | 含义 |
|---|---|
| moment_id | 稳定唯一标识 |
| scenario | A/B/C |
| country_iso3 | country/connector |
| analysis_sector | sector |
| moment_name | 名称 |
| moment_value | 数值 |
| unit | 单位 |
| source_name | 来源 |
| sample_start, sample_end | 样本期 |
| formula_version | 公式版本 |
| direct_or_proxy | direct/proxy/allocated/financial |
| can_enter_model | 是否允许进入模型 |
| exclusion_reason | 不允许时原因 |
| validation_status | PASS/FAIL |

# 13. Controls、fixed effects 和 robustness 变量

这些变量用于后续实证，不应与核心 outcome 混合。

| Family | Variables |
|---|---|
| time | year、post2018、post2020、pre/shock/post indicators |
| connector FE | connector_iso3 categorical |
| sector FE | analysis_sector categorical |
| baseline scale | log pre trade level only when positive；pre level raw |
| baseline China dependence | sector China share 2017 |
| baseline exposure | 2017/2018 sector exposure |
| proxy/scope | TWN proxy、BEA autos proxy、medical unsupported |
| coverage | has source flags、missing/suppression/confidential counts |
| small base | pre_zero、small_base、log_defined |
| alternative windows | 2014-2017 vs 2020-2023；exclude 2024；shock-only |
| source bloc | China-only vs extended East Asia |
| policy | rate vs exposure vs tariffed share |

2024 可能同时受政策 review 和数据时间边界影响。所有包含 2024 的结果要有 exclude-2024 robustness。

# 14. 不允许的变量和解释

| 禁止项 | 原因 |
|---|---|
| 把 Census 15-origin 合计命名为 world imports | 分母范围错误 |
| 把 Census quantity zero 当真实数量 | source unavailable |
| 把 suppression/confidential missing 填 0 | 改变官方含义 |
| 把 BEA position + flow + income 相加 | stock/flow/概念不同 |
| 把 FDI increase 命名为 plant entry | financial evidence 不能证明实体进入 |
| 把 trade rerouting flag 命名为 production relocation | 缺少 direct production evidence |
| 把 allocated AMNE VA/exports/imports 称为直接 bilateral observations | 来源口径错误 |
| 把 repeated China-US benchmark 跨 connector 求和 | 机械重复 |
| 把 rate/share/exposure 直接求和到 sector | 应重算分子分母 |
| 给 many-to-many concordance 等权拆分且不披露 | 无经验依据且会伪精细 |
| 用 arbitrary epsilon 让 ratio/log 看似完整 | 隐藏 undefined denominator |
| 将 Scenario C 称为 structurally identified entry margin | 当前模型不是 endogenous fixed-cost entry model |

# 15. Stata 文件、主键和验证合同

## 15.1 计划交付

| DTA | Stata isid | 用途 |
|---|---|---|
| m2_product_treatment.dta | isid hs6 year | policy product panel |
| m2_connector_trade.dta | isid conn_iso3 hs6 year | product-level trade |
| m3_trade_sector_year.dta | isid conn_iso3 ana_sector year | sector-year trade |
| m3_trade_sector_window.dta | isid conn_iso3 ana_sector window_id | pre/shock/post |
| m3_trade_sector_window_comparison.dta | isid conn_iso3 ana_sector | pre/shock/post comparison and post-pre growth |
| m4_production_long.dta | source-specific composite key | production evidence |
| m5_connector_evidence.dta | isid conn_iso3 ana_sector year | integrated interface |
| m6_connector_ranking.dta | isid conn_iso3 ana_sector rank_spec | ranking |
| m6_structural_moments.dta | isid moment_id scenario | model handoff |

## 15.2 Storage types

- monetary levels：double；
- rates/shares/growth：double；
- year：int；
- month/quarter/flags：byte；
- counts：按范围使用 int 或 long；
- ISO3：str3；
- HS6：str6；
- sector/source/status：固定上限字符串，不使用易丢信息的过短类型。

### 15.3 必须运行的 Stata 检查

```
version 18
clear all
set more off

use "data/processed/stata/m3_trade_sector_year.dta", clear
describe
isid conn_iso3 ana_sector year
assert inrange(year, 2014, 2024)
assert cn_us_val >= 0 if !missing(cn_us_val)
assert cn_conn_val >= 0 if !missing(cn_conn_val)
assert ea_conn_val >= 0 if !missing(ea_conn_val)
assert conn_us_val >= 0 if !missing(conn_us_val)
assert inrange(sec_expo, 0, 1) if !missing(sec_expo)
assert qty_ok == 0 if !missing(qty_ok)

count
summarize cn_us_val cn_conn_val ea_conn_val conn_us_val sec_expo
display "DATA_HANDOFF_VALIDATION_SUCCESS"
```

最终 .do 要逐表检查 rows、keys、ranges、missing counts、totals 和 success marker，并把 log 写入项目 F 盘。

## 16. 每个变量完成后的登记模板

```
Canonical name:
Stata name:
Status:
Output file:
Grain and key:
Source fields:
Formula:
Window:
Unit:
Additivity:
Zero rule:
Missing rule:
Negative-value rule:
Suppression/confidentiality rule:
Denominator rule:
Small-base rule:
Winsorization/standardization:
Purpose:
Allowed interpretation:
Prohibited interpretation:
Validation checks:
Validation result:
Output hash:
Reviewer/date:
```

## 17. 当前施工顺序

1. M3.1 已构造 connector-sector-year levels、coverage 和 sector treatment；
2. M3.1 product-to-sector value reconciliation 与确定性检查已通过；
3. M3.2 已构造 complete-window annual means、coverage/policy summaries、post-pre change、DH/log growth 和 mechanical zero/defined flags；
4. 下一步 M3.3 固定 small-base、异常增长和 winsorization rules；
5. M3.4 构造 replacement、source input、exposure overlap 和 co-movement；
6. M3.5 保留所有 connectors/comparators；
7. M3.6 生成窗口、source bloc 和 exposure sensitivity；
8. M4 逐来源建立 production long；
9. M5 逐来源 left join 和 coverage；

10. M6 先 trade ranking，再 production grade，最后 structural moments；
11. M7 导出 Stata、运行 .do、冻结 release。

# 18. 版本记录

| Version | Date | Change |
|---|---|---|
| v0.1.0 | 2026-07-27 | 建立 M2 已验证变量、M3-M6 planned variables、公式、用途、缺失规则、排名方案、structural moment 边界和 Stata handoff 合同。 |
| v0.1.1 | 2026-07-27 | 与完整数据处理审计档案建立双向合同；规定 M3-M7 每步必须同步登记实际输出、validation、hash 和 Stata 接口。 |
| v0.2.0 | 2026-07-28 | 完成 M3.1 变量合同：登记 1,001 行 connector-sector-year 输出、70 个 canonical/Stata aliases、实际公式、flow-specific coverage、17/11 个 treatment denominator 异常、non-additive flags、validation 和确定性输出 hash；M3.2-M3.6 保持 planned。 |
| v0.3.0 | 2026-07-28 | 完成 M3.2 变量合同：登记 273 × 79 long window 和 91 × 142 comparison 两表接口、complete 4/2/5-year annual means、ratio-of-sums coverage、policy summaries、post-pre change/DH/log growth、zero/defined flags、17/11 denominator 异常、79/142 个 Stata aliases、两张确定性输出 hash 和 M3.3 边界。 |

下一次强制更新：M3.3 完成时，补齐 small-base threshold、异常增长判定、winsorization/capping 规则、敏感性和实际 output/validation/hash；同时更新审计档案并重建两份 PDF。